ZERO AND SURJECTIVITY FUNCTORS OF A MAP VERSUS ITS AFFINE ADJOINT, WITH THE CLOSED IMMERSION CASE

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-flat.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: base change stable adjunction between v and u, vanishing equivalence for adjoint pairs, and a unit factorization argument transferring surjectivity through the exact fully faithful direct image of a closed immersion

Editorial note. The label lemma-relate-zero-affine names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- The situation fixing a map of quasi-coherent modules on an algebraic space over B and the functors on Sch/B attached to a property of the base changed map
- Commutation of pullbacks with base change, and commutation of the direct image along an affine morphism with base change
- The adjunction between pullback and direct image for an affine morphism
- Exactness and full faithfulness of the direct image functor along a closed immersion on quasi-coherent modules

Source reference. Original reference 05MH: sources/stacks-spaces-flat/pinned-original-source.tex, source lines 1233–1233. Prerequisite pointer invoked at proof line(s) 1437; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 07U8: sources/stacks-chow/tags.txt, source lines 9963–9963. Prerequisite pointer invoked at proof line(s) 1455; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 04CJ: sources/stacks-chow/tags.txt, source lines 5693–5693. Prerequisite pointer invoked at proof line(s) 1467; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *In Situation 05MH suppose given an affine morphism $i : Z \to X$ and a quasi-coherent $\mathcal{O}_Z$-module $\mathcal{H}$ such that $\mathcal{G} = i_*\mathcal{H}$. Let $v : i^*\mathcal{F} \to \mathcal{H}$ be the map adjoint to u. Then*

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(1) $F_{v,zero} = F_{u,zero}$, *and*
(2) *if* i *is a closed immersion, then* $F_{v,surj} = F_{u,surj}$.

Proof. Let T be a scheme over B. Denote $i_T : Z_T \to X_T$ the base change of i and $\mathcal{H}_T$ the pullback of $\mathcal{H}$ to Z_T. Observe that $(i^*\mathcal{F})_T = i_T^*\mathcal{F}_T$ and $i_{T,*}\mathcal{H}_T = (i_*\mathcal{H})_T$. The first statement follows from commutativity of pullbacks and the second from Cohomology of Spaces, Lemma 07U8. Hence we see that u_T and v_T are adjoint maps as well. Thus $u_T = 0$ if and only if $v_T = 0$. This proves (1). In case (2) we see that u_T is surjective if and only if v_T is surjective because u_T factors as

$$\mathcal{F}_T \to i_{T,*}i_T^*\mathcal{F}_T \xrightarrow{i_{T,*}v_T} i_{T,*}\mathcal{H}_T$$

and the fact that $i_{T,*}$ is an exact functor fully faithfully embedding the category of quasi-coherent modules on Z_T into the category of quasi-coherent $\mathcal{O}_{X_T}$-modules. See Morphisms of Spaces, Lemma 04CJ. □